

Interactive Edge Orientation Identification: An Optimization Approach

Fabian Damken¹ Wouter M. Koolen^{2,1} Rianne de Heide^{1,2}
¹University of Twente, Enschede ²Centrum Wiskunde & Informatica, Amsterdam

Multi-Armed Bandit Notation

- ▶ $\nu = (\nu_i \mid i \in [K])$ is a K -armed bandit
- ▶ reward distribution ν_i for arm $i \in [K]$
- ▶ reward mean $\mu_i = \mathbb{E}_{X \sim \nu_i}[X]$
- ▶ in each round t , agent selects an arm $I_t \in [K]$
- ▶ subsequently receives reward $X_t \sim \nu_{I_t}$

Pure Exploration

- ▶ find out certain details about the bandit
- ▶ e.g., identify best arm $\operatorname{argmax}_{i \in [K]} \mu_i$
- ▶ algorithm components: sampling rule $(\pi_t)_t$, stopping rule τ , decision rule $\hat{m}_\tau$
- ▶ fixed-budget: fix τ , max. $\mathbb{P}(\hat{m}_\tau = m(\nu))$
- ▶ fixed-confidence: fix $\mathbb{P}(\hat{m}_\tau = m(\nu))$, min. τ

Anytime-Valid CIs

- ▶ confidence intervals (CIs):

$$\forall t \in \mathbb{N}: \mathbb{P}(\mu \in \hat{\mu}(t) \pm \beta(t)) \geq 1 - \alpha$$
- ▶ only for *fixed* sample size t
- ▶ does *not* hold if t depends on CI itself
- ▶ anytime-valid CIs hold for all *stopping* times!

$$\forall \tau: \mathbb{P}(\mu \in \hat{\mu}(\tau) \pm \beta(\tau)) \geq 1 - \alpha$$
- ▶ naive: $\alpha = 6\tilde{\alpha}/(\pi^2 t^2)$, better versions exist [1]

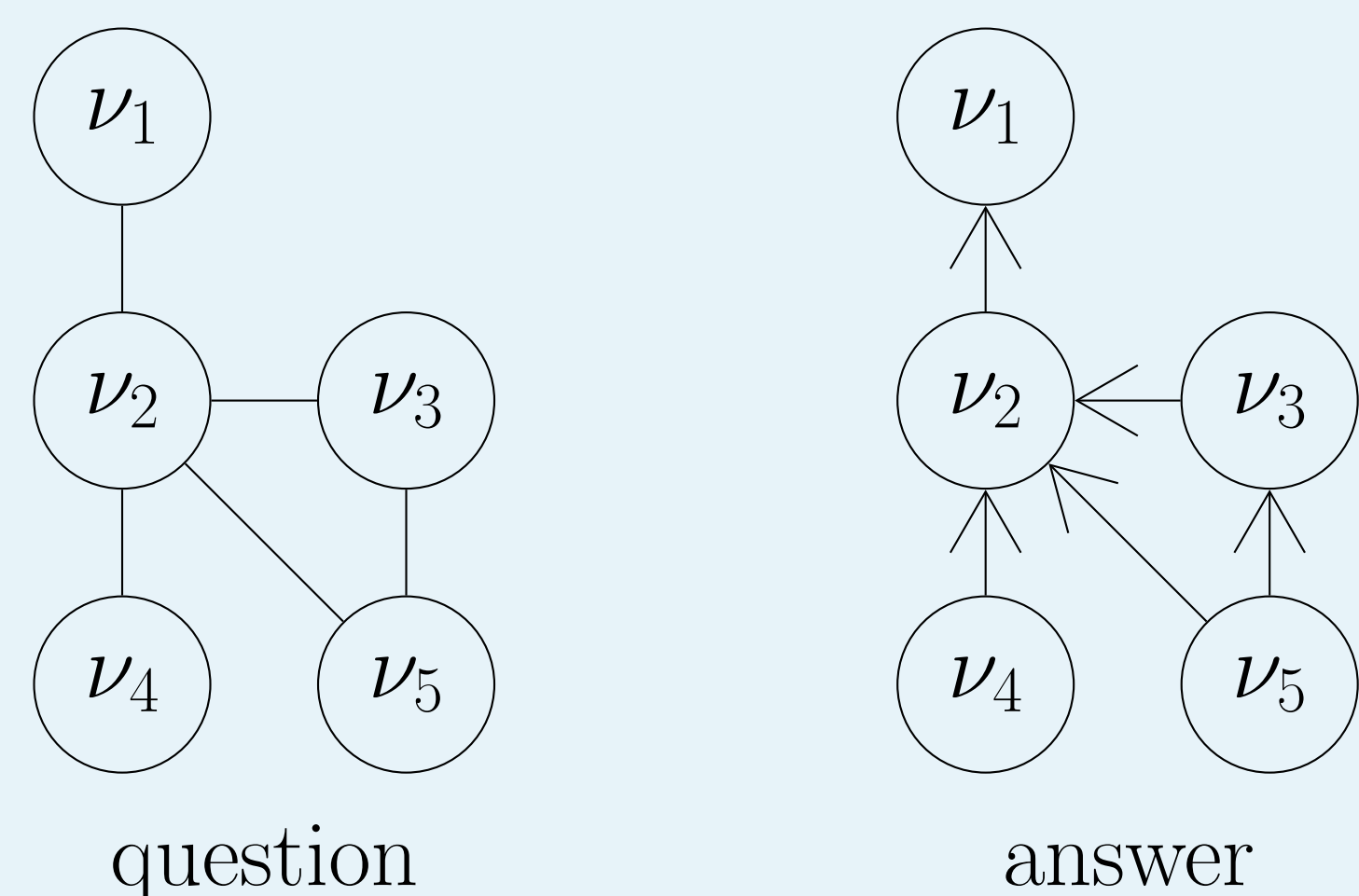
References

- [1] S. R. Howard et al. "Time-Uniform, Nonparametric, Nonasymptotic Confidence Sequences." In: *The Annals of Statistics* 49.2 (2021), pp. 1055–1080. ISSN: 0090-5364. JSTOR: 27169410.
- [2] A. Garivier and E. Kaufmann. "Optimal Best Arm Identification with Fixed Confidence." In: *Conference on Learning Theory*. Conference on Learning Theory. PMLR, June 6, 2016, pp. 998–1027.
- [3] H. Zhuang, C. Wang, and Y. Wang. "Identifying Outlier Arms in Multi-Armed Bandit." In: *Advances in Neural Information Processing Systems*. Vol. 30. Curran Associates, Inc., 2017.

Edge Orientation Problem

consider following 5-armed bandit and task:

- ▶ consider a 5-armed bandit with
 $\mu_1 < \mu_2 < \dots < \mu_5$
- ▶ task definition: graph over arms



task: identify edge orientation

Formal Definition

- ▶ given a set of pairwise queries

$$\mathcal{C} \subseteq \{(i, j) \mid 1 \leq i < j \leq K\}$$
- ▶ we want to decide whether $\mu_i < \mu_j$
- ▶ correct answer:

$$m_{(i,j)}^*(\nu) = (-1)^{\mathbb{1}[\mu_i < \mu_j]}$$

i.e., $m_{(i,j)}(\nu) < 0$ iff $\mu_i < \mu_j$
- ▶ track-and-stop [2] lower bound

Separation Lemma [cf., 3]

- ▶ consider a confidence intervals $k \in \{1, 2\}$:

$$\mathbb{P}(\mu_k \in \hat{\mu}_k \pm \beta_k) \geq 1 - \alpha$$
- ▶ conditional on the CI holding:

$$2(\beta_1 + \beta_2) \leq \Delta \implies \beta_1 + \beta_2 \leq \hat{\Delta}$$
- ▶ i.e., go from true to estimated mean gap

Core Idea

- ▶ place anytime-valid CIs on all arms:

$$\hat{\mu}_i(T) \pm \beta_i(T) \quad \beta_i(T) = \frac{\gamma(T)}{\sqrt{N_i(T)}}$$
- ▶ decide for each $(i, j) \in \mathcal{C}$ whether
 $\mu_i < \mu_j$ based on these CIs
- ▶ directly optimize for stopping time:

$$\min_{N \in \mathbb{N}^K} \sum N$$

s.t. $\gamma\left(\sum N\right) \left(\frac{1}{\sqrt{N_i}} + \frac{1}{\sqrt{N_j}}\right) \leq \Delta_{ij}$

where $\Delta_{ij} = |\mu_i - \mu_j|$

Reformulation

1. relax optimization problem into continuity
2. eliminate $\gamma(\sum N)$ from constraints and objective without changing the argmin
3. replace individual number of pulls $(N_i)_{i \in [K]}$ with total number of pulls T and sampling distribution $\omega \in \Delta_K$
4. massage constraints into the form $\text{LHS} \leq T$
5. replace objective with maximum over LHS's
6. take reciprocal of objective

$$\max_{\omega \in \Delta^K} \min_{(i,j) \in \mathcal{C}} \Delta_{ij}^2 \left(\frac{1}{\sqrt{\omega_i}} + \frac{1}{\sqrt{\omega_j}} \right)^{-2}$$

Algorithm Ingredients

1. *estimate* $\Delta_{ij}^2 \approx \hat{\Delta}_{ij}(t)^2 = |\hat{\mu}_i(t) - \hat{\mu}_j(t)|^2$
2. *optimize* above objective w.r.t. ω , yielding sequence $(\omega(t))_t$, once GD step per iteration
3. *track* cumulative distribution $\sum_{s=1}^t \omega(s)$
4. *stop* if confidence intervals are separated

Upper Bound

- ▶ from the above separation lemma
- ▶ and optimization problem reformulation
- ▶ algorithm with pull distribution ω stops with high probability after T pulls if

$$\frac{1}{T} \leq \frac{1}{4\gamma(T)^2} \min_{(i,j) \in \mathcal{C}} \underbrace{\Delta_{ij}^2 \left(\frac{1}{\sqrt{\omega_i}} + \frac{1}{\sqrt{\omega_j}} \right)^{-2}}_{f^*(\omega) = \quad / \quad f^{(t)} = (\Delta_{ij} \rightarrow \hat{\Delta}_{ij}(t))}^{-2}$$
- ▶ i.e., $4\gamma(T)^2 \leq T f^*(N(T)/T)$
- ▶ lower-bound RHS with reality:

$$\begin{aligned} & T f^*(N(T)/T) \\ & \geq T f^*\left(\sum_{t=1}^T \omega(t)\right) - \mathcal{O}(1) \\ & \geq \sum_{t=1}^T f^*(\omega(t)) - \mathcal{O}(1) \\ & \geq \sum_{t=1}^T f^{(t)}(\omega(t)) - \mathcal{O}(\sqrt{T}) \\ & \geq \sum_{t=1}^T f^{(t)}(\omega^*) - \mathcal{O}(\sqrt{T}) \\ & \geq T f^*(\omega^*) - \mathcal{O}(\sqrt{T}) \end{aligned}$$

where $\omega^* = \operatorname{argmax}_{\omega \in \Delta_K} f^*(\omega)$
- ▶ hence, our algorithm stops with high probability after T rounds if

$$4\gamma(T)^2 + \mathcal{O}(\sqrt{T}) \leq T f^*(\omega^*)$$

Proposed Algorithm

```

1 initialize  $\hat{\mu}_i(0) \sim \nu_i$ ,  $\omega_i(0) = 1/K$ ,  $N_i(0) = 1$ 
2 foreach  $t = 1, 2, \dots, \tau$  do
3    $\hat{\Delta}_{ij}(t-1) = |\hat{\mu}_i(t-1) - \hat{\mu}_j(t-1)|$ 
4    $G^{(t)} = (\nabla_{\omega} f^{(t-1)}(\omega))|_{\omega=\omega(t-1)}$ 
5    $\tilde{\omega}(t) = \omega(t-1) + \eta_t G^{(t)}$ 
6    $\omega(t) = \Pi_{\Delta_K^e} \tilde{\omega}(t)$ 
7    $P(t) = \sum_{s=1}^t \omega(s)$ 
8   pull  $i_t \in \operatorname{argmax}_{i \in [K]} [P_i(t) - N_i(t-1)]$ 
9   if  $\frac{\gamma(t)}{\sqrt{N_i(t)}} + \frac{\gamma(t)}{\sqrt{N_j(t)}} \leq \hat{\Delta}_{ij}(t)$  then
10    set  $\tau = t$ 
11  end
12 end
13 return  $(\hat{m}_\tau)_{(i,j)} = (-1)^{\mathbb{1}[\hat{\mu}_i(\tau) < \hat{\mu}_j(\tau)]}$ 

```